

Regular Quiz 02

8 question(s) — correct option(s) marked in green.

Q1. When the child process reaches Line 7, for which memory segment(s) was a Copy-on-Write (COW) triggered?

```
1. int var1, var2=100;
2. int main() {
3.     if(fork() == 0) {
4.         var1 = ++var2;
5.     }
6.     printf("Var1=%d", var1);
7.     return 0;
8. }
```

- A. .data (correct)
- B. .bss (correct)
- C. .rodata
- D. heap

Q2. Recall Lecture 05 slide on Processes P1 and P2 arriving at t=0 on a uniprocessor system. P1 enters the ready queue first. What is the total execution time for both processes? Consider no other processes exists.

P1: 4ms CPU -> 6ms I/O -> 4ms CPU
P2: 2ms CPU -> 1ms I/O -> 3ms CPU

- A. 14ms (correct)
- B. 13ms
- C. 15ms
- D. 20ms

Q3. When a trap occurs, the CPU saves the user registers — onto which stack?

- A. The user stack, since they're "user" registers
- B. At the bottom of kernel stack (correct)
- C. They just stay in the registers, never pushed anywhere
- D. At the top of kernel stack

Q4. A child calls `exit(7)` a full second before the parent ever calls `wait(&ret)`. What happens when the parent finally calls `wait()`?

- A. It blocks forever
- B. It returns immediately (correct)
- C. It returns -1 (error)
- D. The exit status is lost by then; ret ends up garbage

Q5. What should be the optimal size of shared memory segment for parallelizing the below code among 2 processes?

```
int A[10], B[10], C[10]; //all initialized
...
for(int i=0; i<10; i++) {
    C[i] = A[i] + B[i];
}
```

- A. 120 bytes
- B. 40 bytes (correct)
- C. 80 bytes
- D. 44 bytes

Q6. Select all TRUE statements about POSIX semaphores

- A. sem_wait() decrements first, blocks only if the result is negative (correct)**
- B. sem_post() blocks until the woken process resumes
- C. sem_post() always wakes waiters in strict FIFO order
- D. A negative value means processes are currently blocked (correct)**

Q7. Semaphores "permit" and "mutex" initialized with 3 and 1, respectively. When 3 processes execute the following codeblock simultaneously, how many can be in CS at once, and how many are blocked on "permits"?

```
sem_wait(&permits);  
sem_wait(&mutex);  
/* Critical Section (CS) */  
sem_post(&mutex);  
sem_post(&permits);
```

- A. 1 in CS, 0 blocked (correct)**
- B. 3 in CS, 0 blocked
- C. 1 in CS, 2 blocked
- D. 3 in CS, 1 blocked

Q8. Order the following IPC mechanisms by increasing system overhead and latency per byte transferred (without considering the setup cost)

- A. Signals -> Pipes -> Shared memory
- B. Pipes-> Signals ->Shared memory
- C. Shared memory-> Pipes -> Signals (correct)**
- D. Signals -> Shared memory -> Pipes